

Natural flavors are derived from natural sources—plants, animals, spices, fruits, vegetables, roots, or fermentation products—and are isolated through physical, enzymatic, or microbiological processes, according to FDA regulations. These processes include extraction, distillation, and fermentation, which concentrate flavor compounds. 🍷 Rip Van +2

Here are the key sources and methods based on industry documentation:

## 1. Extraction Methods

Extraction uses solvents to dissolve and pull out flavor compounds from natural sources. 🍷 Rip Van

- **Solvent Extraction:** Involves soaking natural materials in water, alcohol, or other solvents to create concentrated extracts, such as vanilla extract from vanilla beans.
- **Cold Pressing:** Primarily used for citrus fruits, this mechanical process crushes peels to release essential oils.
- **Enzymatic Extraction:** Enzymes (e.g.,  $\beta$ -glucosidase) are used to develop intense aromas from sources like cherry or almond kernels.
- **Supercritical Fluid Extraction:** A modern, eco-friendly method, often using  $CO_2$ , to extract flavors without leaving residual solvents.
- **Sonication:** Uses sound pressure to disrupt cell walls and increase the release of flavor components. 🍷 Rip Van +5

## 2. Distillation Methods

Distillation separates compounds based on boiling points to isolate volatile aromatic compounds. 🍷 Rip Van

- **Steam/Hydro Distillation:** Plant material is heated with steam, causing essential oils to release and then condense into a concentrated form (e.g., peppermint oil).
- **Fractional Distillation:** Used to further isolate specific components from a raw extract, such as isolating vanillin from vanilla extract. 🍷 Rip Van +2

## 3. Fermentation & Biological Processes

Microorganisms are used to transform raw ingredients into complex flavor compounds. 🍷 Rip Van

- **Microbial Fermentation:** Bacteria, yeast, or fungi are used to convert carbohydrates, lipids, and proteins into, for example, dairy-like, fruity, or nutty flavors.
- **Roasting/Heating:** Natural materials are heated to create complex, savory, or sweet flavor profiles (e.g., cocoa or coffee extracts). 📖 PubMed Central (PMC) (.gov) +4

## Natural Source Material Categories

Natural flavors are derived from the following sources, as outlined in USDA and FDA regulations: 🌐

- **Botanicals:** Fruits, vegetables, herbs, roots, bark, and buds.
- **Animal Products:** Meat, seafood, poultry, eggs, and dairy products.
- **Microorganisms:** Edible yeast and other fermenting agents. 🌱 Beck Flavors +4

For more specific information on the legal definitions, you can consult the [USDA Policy Memo 11-1 on Natural Flavors](#). 🌐